# Unit IV

# Physics of Nanoparticles

# Nanoparticles

## Introduction to nanoparticles and nanotechnology :-

Nanotechnology is science, engineering, and technology conducted at the nanoscale, which is about 1 to 100 nanometers. It is about the study and manipulating matter on an atomic and molecular scale.

## Quantum confinement, surface to volume ratio and properties :-

**Quantum confinement:**

- A bulk material or solid has three dimensions.
- If only one length of three- dimensional nanostructure is in nanoscale (1-100 nm), the structure is known as a quantum well.
- Instead if two sides are of nanoscale (1-100 nm), the resulting structure is known as quantum wire.
- A nanoparticle is often referred to as a quantum dot as its all three dimensions are in the nanoscale (1-100 nm).

**Surface to volume ratio:**

- As the size of materials approaches nanoscale, the percentage of atoms at the surface of a material becomes significant.
- Nanoparticles have a very high surface area to volume ratio.

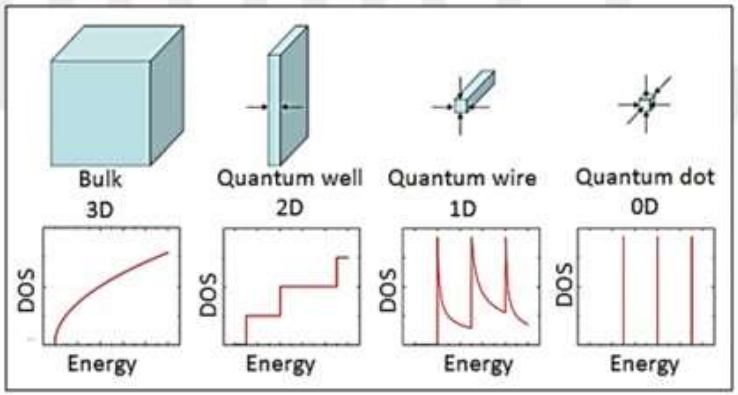



*Figure: Quantum confinement effects*

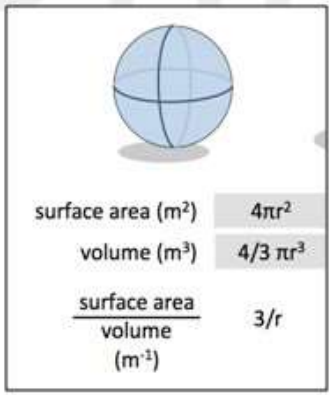



*Figure: Surface to volume ratio*

## Significance of nanoscale and change in properties of nanoparticles:

Many properties of solids such as optical, electrical, mechanical, etc depend on its size. As compared to its bulk form, properties of materials change when at least one of the dimensions of material is reduced to naoscale (1-100 nm)

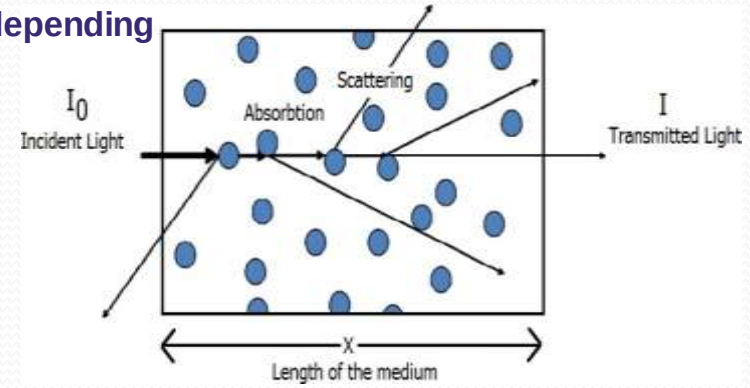
mainly due to following reasons:

- Upon reaching quantum confinement at nanoscale energy levels of nanoparticles become discrete (as shown in the figure). This increases or widens up the band gap and the band gap energy. This results into change in the properties of material at nanoscale.
- Nanoparticles have a relatively larger surface area when compared to its bulk form. This makes the nanoparticles more chemically reactive compared to bulk form and it thus modifies its properties.
- Physical properties of materials such as electrical are characterized by the mean free path (path of the electron collision due to atoms or ions). This mean free path changes at nanoscale.
- Bulk properties of materials such as hardness changes at nanoscale. Nanoparticles are much harder and tougher as compared to their bulk form.

## Optical properties of Nanoparticles :-

The color of nanoparticles is different from bulk material. One of the daily life examples is of glass. Glasses are transparent, but when doped with different nanoparticles, adopt different colors like red, pink, blue, green, etc depending upon the dissolved nanoparticles.

| Metal | Color for bulk | Color at nano-level |
|---|---|---|
| Gold | Yellow | Bright red |
| Silver | Colorless | Pale yellow |




## G. Mie Theory for change in optical properties

- G. Mie explained the phenomenon using Maxwell's equation in 1908.
- When electromagnetic radiation incidents on spherical particles of uniform size a part of the radiation is absorbed and a part is scattered.
- Thus the intensity of transmitted light changes.
- If incident light has wavelength $\lambda$, intensity $I_0$, length of medium $x$ then The intensity of transmitted radiation is given by:

$$I = I_0 e^{-\mu x}$$

Where, $\mu = (N / V) C_{ext}$ is extinction coefficient

N is number of particles in medium, V- Volume of colloidal particles

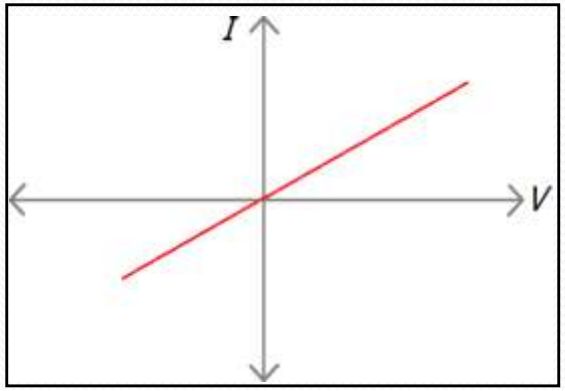
$C_{ext}$ – extinction cross section of a particle

- At nanoscale, as size of the nanoparticles change, there is change in the length of medium and thus the values of $\mu$ and $x$ changes.
- This changes the intensity of transmitted light and it depends on the wavelength of incident light.
- This changes the color of nanoparticles at different nanoscale region.
- Mie theory successfully explained absorption of light of visible wavelength for nano- particles,
- but for particle size less than ~10 nm, there is need to consider dielectric constant which depends also on particle size.

# Electrical properties of Nanoparticles

•Electrical properties are characterized by the mean free path of electron (path of the electron collision due to atoms or ions).

•The mean free path changes at nanoscale due to collision and tunneling within the conductor.




**I-V Characteristics of bulk conductor:**

When voltage *V* is applied across the conductor, current *I* flowing through it is governed by Ohm's Law and has a linear variation.

**I-V Characteristics at nanoscale:**

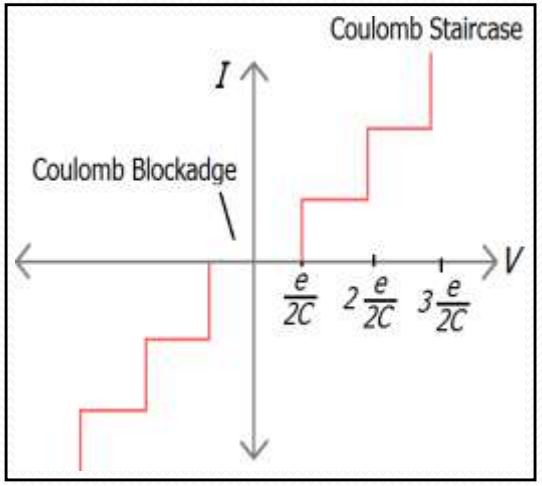
•If dimensions of conductor are in nanoscale (1-100 nm), there appears a region around zero voltage for which there is no current.

•A single electron is transferred by tunneling when the voltage is ± *e*/2*C*. Therefore, when the voltage is less than this electron cannot be transferred.

•This gives a region of zero current at low bias voltage and is known as 'Coulomb blockade region'.

•The electrons are transferred when the voltage is ± *e*/2*C*.

•The repeated tunneling of single electron produces 'Coulomb staircase'.




**Electrical resistivity:**

•The materials of nano-sized grains have larger number of grain boundaries than corresponding polycrystalline materials.

•Electrical resistivity of materials having nano-sized grain is larger than the polycrystalline materials due to scattering of electrons at grain boundaries.

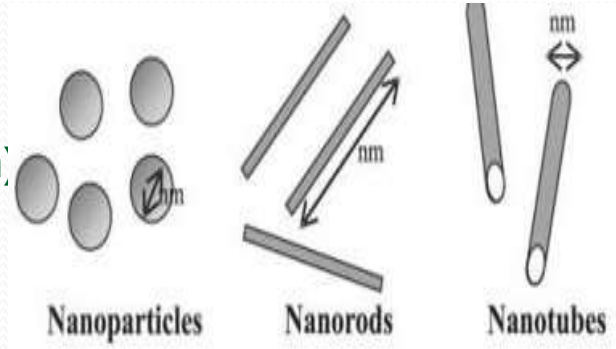
# Mechanical properties of Nanoparticles

•Mechanical properties of materials depend upon the composition of material, bonds between the atoms and presence of impurity.

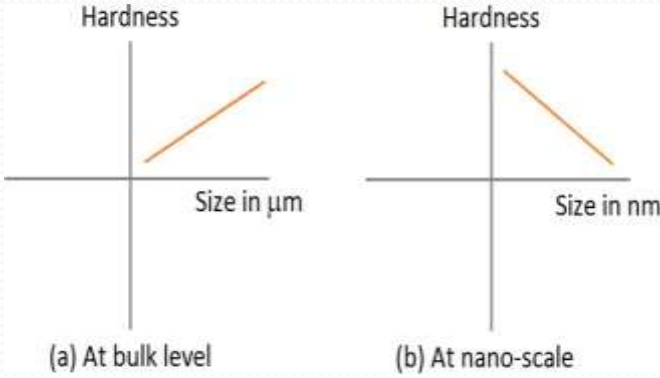
• When the size of materials is reduced to nano-scale (1-100 nm) materials tend to be single crystal.

• The measurements of elasticity, hardness, ductility show different behavior as compared to the bulk material.

•For example, in the case of metallic nanocrystalline materials, elastic modulii reduce dramatically.




| Material | Form | Grain size | Young's modulus |
|---|---|---|---|
| Mg | Nanocrystal | ~12 nm | Y = 3900 N/mm$^2$ |
| | Polycrystalline | > 1mm | Y = 4100 N/mm$^2$ |

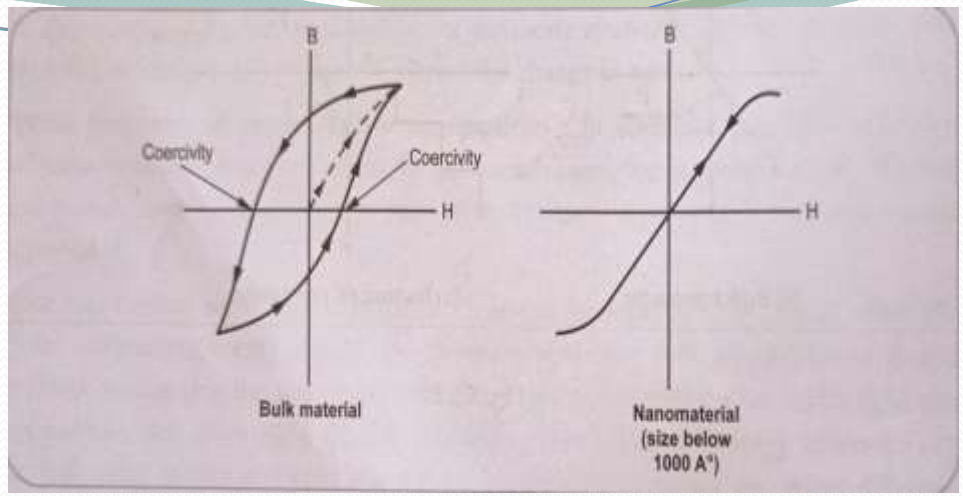


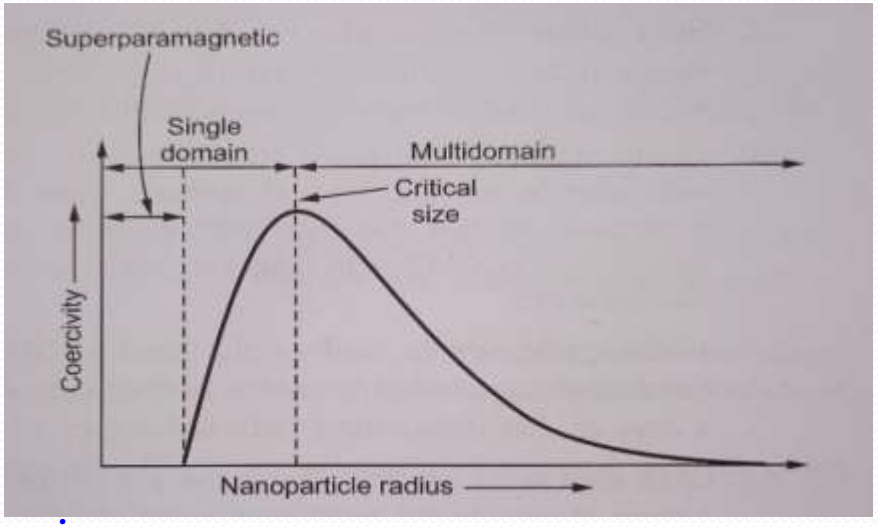

•Plastic deformation in nanocrystalline materials strongly differ from that of polycrystalline bulk material i.e. if stress is removed the recovery to original shape/size is recovered more effectively.

• Hardness of materials is also related to the grain size. In the nanometer size range, the hardness increases with decrease of particle size. (Shown in figure for copper)

# Magnetic properties of Nanoparticles

- Ferromagnetic bulk material converted to super paramagnetic when the particle size is reduced. This happens due to change in surface area.
- When size of ferromagnetic Material reduced below a critical value, it becomes a single domain. When size of single domain particle further decreases below a critical diameter, the coercivity becomes zero and such particles become super paramagnetic.
- It has been found that as the particle size is reduced, the coercivity increase to maximum and then decreases towards zero.



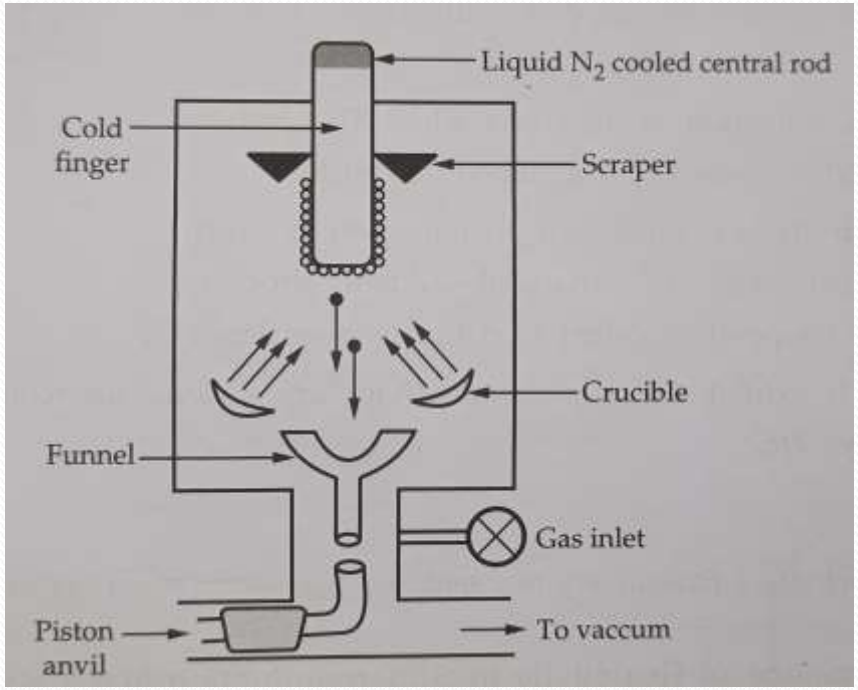




Some of the possibilities when we go from bulk to nano are:

- Ferromagnetic particles becoming single domain.
- Superparamagnetism in small ferromagnetic particles (i.e. particles which are ferromagnetic in bulk)
- Antiferromagnetic particles (in bulk) behaving like ferromagnets etc.
- There is a increase in magnetic moment/atom as we decrease the dimensionality of the system.

# Synthesis of Nanoparticles by Physical Vapour Deposition (PVD)

- Physical vapor deposition (PVD) is a dynamic bottom-up synthesis technique capable of producing thin film materials at a nanoscale (1-100nm). In physical vapour deposition, the material is heated above its melting point to generate vapours, which are then deposited on the target surface.
- Thermal evaporation involves heating a source above its melting point in order to extract atoms from it.
- Typical evaporation method consists of material for evaporation, inert gas, cold finger and scrapper. Material for evaporation is kept in crucible, condensation of nanoparticles takes place at cold finger, scrapper is to scrap the the nanoparticles which are condensed on cold finger.




- The physical vapour deposition (PVD) method is used to deposit any type of inorganic materials such as metals, alloys, and organic materials.
- PVD takes place in a vacuum condition.
- Using physical vapour deposition techniques, However, it is challenging to deposit a consistent combination of two distinct materials because of the variation in melting. Both solids and liquids can evaporate and produce vapours. The solid particles are produced by the nucleation and condensation of the saturated vapour

## •Advantages

- **Environmentally friendly: PVD is an environmentally friendly method that doesn't cause pollution.**
- **Thin films: PVD can produce thin films of nanoparticles.**
- **Crystal structure: Nanoparticles produced by PVD have a good crystal structure.**
- **Control: The size of the nanoparticles can be controlled.**

## • Disadvantages

- **Cost: PVD is expensive to produce.**
- **Productivity: PVD has low productivity.**
- **Equipment: PVD requires complex equipment.**
- **Speed: PVD is slower than other coating deposition processes.**
- **Substrates: PVD is limited to substrates with simple geometries.**

## •Applications-

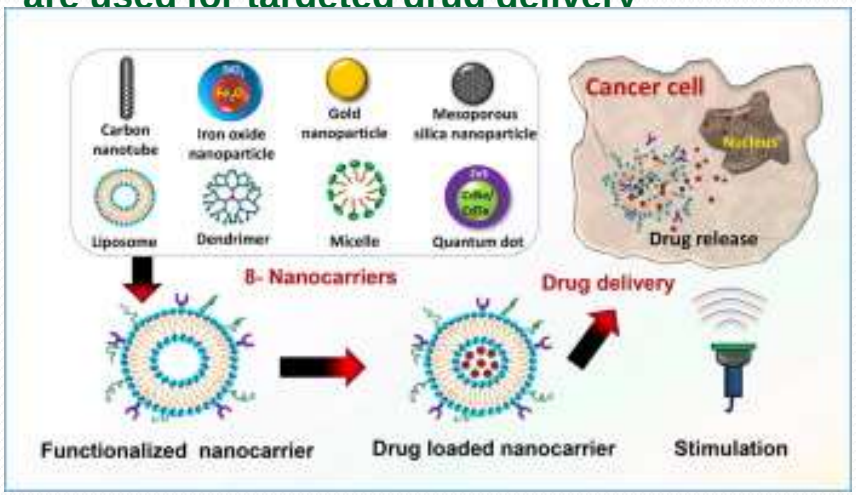
- **Optical interference coatings**
- **Mirror coatings**
- **Decorative coatings**
- **Permission barrier films**
- **Electrically conducting films**
- **Corrosion protective coatings**

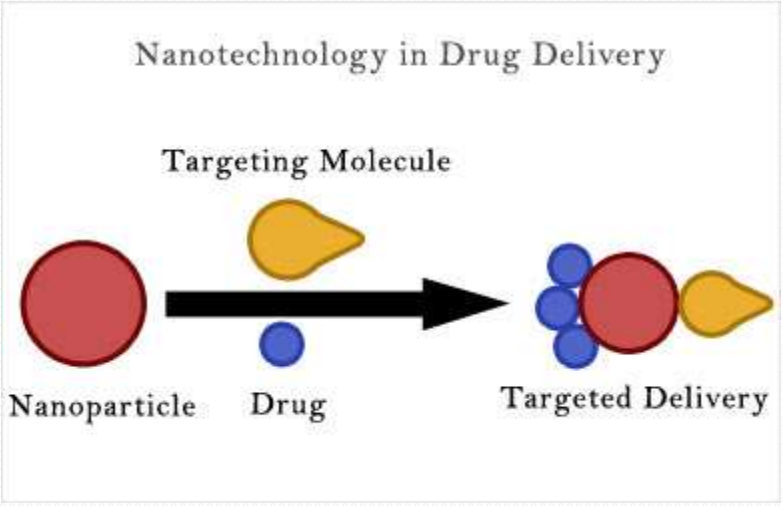
# Applications in medicine (targeted drug delivery)

- The traditional drug delivery system, the drug is distributed to the affected tissues or organs of the patient's body as well as to the healthy tissues or organs. This leads to the side effects.
- In the targeted drug delivery system, the drug is dissolved, entrapped, encapsulated or attached to a nano particle. The system is then embedded in a capsule which is guided towards the affected part of the body.
- The capsule is opening at the specific tissues or organs controlled by externally applied magnetic field or infrared light or physiologically. Then drug can be delivered in controlled manner.

**Carriers for targeting drug**

**Depending on the treatment required and nature of the drug, following are few examples of carriers are used for targeted drug delivery**




| Sr. | Carriers of drug | Applications |
|---|---|---|
| 1 | Poly (alkyl cyanoacrylate) nanoparticles with anti cancer agents | Cancer therapy |
| 2 | Poly (methyl methacrylate) nanoparticles with vaccines | Vaccine delivery |
| 3 | Poly(alkyl cyanoacrylate) polyester nanoparticles with anti parasitic | Intracellular targeting |
| 4 | DNA- gelatin nanoparticles | DNA delivery |

# Applications of nanotechnology in Electronics

In the spin-electronic devices, the spin property of electrons (instead of charge) is taken into consideration. Advantage with spin is that it cannot be easily destroyed by scattering from collisions with other charges, impurities or defects. The electronic devices with typical dimensions of few nanometers in either of three directions display unique properties. Such devices are fast, compact, and relatively cheap. Such devices are typically known as spin-electronic (spin-electronic) devices. Few applications are listed as below:

***Single electron transistor (SET) :-***The single electron transistor is a switching device in which a quantum dot is placed between the source and the drain. It uses controlled electron tunneling to amplify current. They offer low power consumption and high operating speed.

***Spin valves:-***Spin values are commercially used in computer read heads. Their use has enabled to increase the data storage capacity of magnetic memory devices due to their ability to detect small magnetic fields.

***Giant Magneto Resistance (GMR):-***GMR effect is the change in magnetic field of certain materials after application of magnetic field. This is very effective in observing small changes in the magnetic field and useful as a read device of the magnetically stored data. The data storing and reading in computer hard disks is the direct application of GMR.

***Spin polarized Field Effect Transistor (S-FET):-***In S-FET Polarized beam of light was used to obtain spin polarized electrons from the source. It is expected that this will help in faster and efficient data processing.

***Nanophotonics:-***In nanophotonics, nanostructures (quantum dots, nanowires or 2-D thin films) or nanocomposites are used to produce light or detect light. They can be used for light production, propagation, manipulation like amplification, filter, detection etc.

***Display technologies for TV and monitor:-***The flat panel television or computer monitors are products of nanotechnology. Even the coatings used on screens of TV or monitors can be of nanoparticles, which have better properties in terms of color quality and resolution than micro particle coatings.

# Applications of nanotechnology in Space and defense

Nanotechnology has a wide range of applications in space defense. Nanomaterials in various forms are used for reduction of weights of space vehicles, improving efficiency of solar cell, insulation of space vehicle, ignitior and propellents, etc. A few of them are listed below:

***Aerogels:-***

Aerogels are designed using nanomaterials and have very low density typically 0.01–0.8 g/cm$^3$. They are very light weight and poor conductor of heat and have potential applications in designing spacecrafts, defense equipment, suits, jackets to reduce the weight.

***High efficiency light weight solar cells:-***

Satellites or spacecrafts are mainly powered by solar energy. Currently solar cells have reached an efficiency of 30 - 40 %. Using luminescent dye sensitized nanoparticle based or nanoparticle-based solar cell arrays have potential to reduce the weight of solar cells as well as increase in efficiency.

***Insulation for space vehicles:-***

Space vehicles should withstand harsh and extreme environments during launching and in space. Materials should also sustain high or low temperature and high or low pressure. Use of silica fibers and nanoparticles provide insulation in solid rocket motors and also they are better radiation protectors.

***Better ignitors and propellants:-***

Nanocrystalline materials such as alumina particles are better propellants than conventional one. A nanocomposite of $Fe_2O_3$ and aluminium burns much faster and is more sensitive than conventional thermites.

***Fatigue resistant materials:-***

Fatigue strength in aircraft usually decreases with time. Some nanomaterials have better fatigue strength and life is increased by 200–300 %.

***Detection of biological weapons:-***

Biological weapons may use dangerous microbes or viruses as weapons. Some nanoparticle oxides like CaO, $Al_2O_3$ and MgO interact with such chemicals much faster than microparticles and are ideally suited for fast decomposition of warfare chemicals

# Applications of nanotechnology in Automobiles

**Nanotechnology has a wide range of applications in automobiles. Nanomaterials in various forms are used for designing sturdy structural parts, smooth and non-scratch paints, self-cleaning windows, better tires, reduction of harmful emission, etc. A few of them are listed below:**

***Sturdy structural parts :-*Various body parts of vehicles are made up of steel, some alloys, rubbers, plastics etc. Nanotube composites have mechanical strength better than steel.**

***Smooth and non-scratch Paints :-*Cars are spray painted with fine particles. Nanoparticle paints provide smooth, non-scratch thin attractive coating.**

***Self cleaning glass for windows:-*Self cleaning glass can be made by dissolving small amount of titanium ($TiO_2$) nanoparticles while manufacturing glass. Titanium is able to dissociate organic dust in presence of UV light available in the sunlight. Drops of waters on glass give hazy look, but $TiO_2$-containing glass can spread water evenly giving clear sight.**

***Small motor parts:-*Small motors are needed in vehicle (such as wipers, window glass movements, etc). Very powerful electric motors are made using shape memory alloys using nanoparticles of materials like Ni–Ti. They require less power and gives better performance.**

***Better tires:-*Tires of vehicles undergo wear and tear with use and also increase its weight. By using nanoparticle clay, better, light weight, less rubber consuming thinner tires are possible that can reduce the overall weight, increase in fuel consumption and speed.**

***Controlling harmful emission:-*Use of efficient nanomaterial catalysts is one solution to convert harmful emission into less harmful gases. Large surface area of nanoparticles is useful to produce better catalysts. They are capable of absorbing emission of particles and poisonous gases like CO and NO from vehicle exhausts.**

***Hydrogen fuel:-*Instead of using conventional petro fuels, there are numerous advantages of using hydrogen as a fuel. When hydrogen fuel is burned it can only produce harmless water vapor. Hydrogen gas is normally stored in a metal cylinder under high pressure. Nanocylinders of carbon nanotubes have potential to store hydrogen in very small space.**